



CargoFish: A "Physical Internet" delivery utility! (Funded from August 2024 to November 2025 by a \$75k Clean Tech Seed Grant from the New Jersey Commission on Science Innovation and Technology - THANK YOU!) CargoFish conducted a public proof-of-concept prototype demonstration at the NYC Fleet Show, Corona Flushing Meadows Park, Queens, NY on May 15th, 2025, and a more elaborate prototype demonstration from August 4th



...links, or visit our [page](#) [explaining environmental issues](#) (scroll down to the bottom of about page for main explanatory video, too.)

Move Only What Matters!

The physical internet for eco-friendly, cost-effective movement of Consumer Packaged Goods

Each month, American households stay supplied through an average of:

...305 million deliveries

...7 billion shopping excursions

Results:

...Billions of gallons of diesel and gasoline burned to fuel those delivery trucks and cars

...Roads in constant need of repair

...Traffic congestion

...Accidents, injuries and fatalities

...Millions of tons of waste generated by packaging materials

As online shopping continues to explode, these issues will just get worse. The root cause?

Standard shipping methods are simply not equipped to accommodate vast quantities of small packages delivered on a regular basis.

And forget about the large scale issues - current shipping methods are also really inconvenient for consumers.



We're used to it, so we think it's normal. But let's put it in perspective. Would you ever consider trekking to the well for your daily drinking water? Or heating your home with a fresh supply of coal delivered weekly?

Of course not. Because today, water and heat come to you through utilities.

But you know what we still do?

We drag our 3000-pound cars through

2 miles of backed-up roads

using ¼ gallon of gas

to pick up 20 pounds of groceries.

Not a very efficient use of resources.

That's why CargoFish envisions a day when these inefficiencies will be completely eliminated.

How?

With the final utility - otherwise known as the physical internet.

Just as information flows from place to place instantaneously with the click of a button, unseen, physical products will flow from location to location, underground.

Imagine if all water had to be bottled and shipped to households. It would cost 10,000x more. A single liter of bottled water costs around \$2, and produces plastic waste. A liter of water from the tap? It costs less than a penny and generates no waste.



What if we leveraged today's available technologies to add one more network? A

network designed to do for small solid payloads what the Internet did for small packets of information? One more pipe to move packages, the way wires deliver electricity and pipes deliver gas?

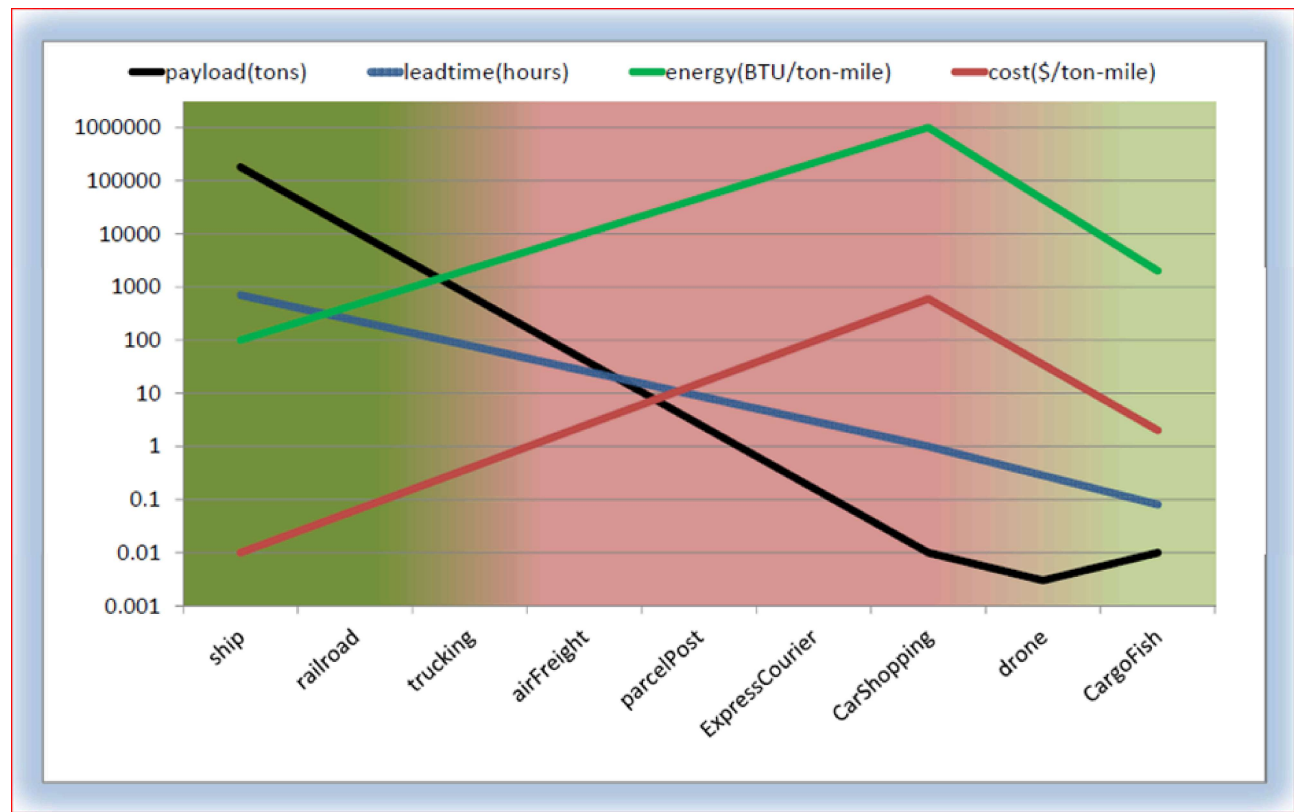
...Overhead expenses for distribution of Consumer Packaged Goods would plummet.

...These savings would be passed on to consumers in the form of cheaper goods!

...Levels of service and convenience would skyrocket

...There would be huge energy savings, and a tremendous decrease in the burning of fossil fuels

...Packaging waste could be slashed - and there'd be no need for plastic grocery bags.



Payload Mode Comparison



Unlike other “trendy” shipping solutions, such as drones, the CargoFish solution is:

...**All-weather** - it is underground, unaffected by storms, and can even be engineered to avoid major damage from earthquakes.

...**High security** - when packages are shipped through pipes directly into a consumers home, there is little chance of theft

...**Low liability** - nor is it likely that the package will cause or be damaged!

...**Low specific cost** - once installed and in use, the cost of each delivery will be minuscule

...**Low specific energy** - since CargoFish system consists of self-propelled vehicles on energized rails, very little energy is used to move each parcel.

The only real expense? Installation. But like all other utilities, the system will be so long-lived and widely used that the installation expense will RAPIDLY be covered by the tremendous savings and profits.

In other words, it will provide the same extreme value that has been demonstrated by every other utility distribution system that has preceded it... *none of which anybody could imagine having a city without.*

“What a game-changing, genius innovation. I can't wait for the day when this delivery system is a reality. Infrastructure always seems daunting, but that's why the sophistication of a country is often measured by its infrastructure. What makes a nation 'developed' anyway?” - Sandy Zylka, VP Products & Technology at NextAxiom Technology

US households consume \$5+ trillion annually in fast-moving consumer goods.

The need is tremendous. And the solution is here.



CargoFish is like a miniature subway system with cylinder-shaped tunnels of an 8-inch diameter.

The infrastructure is comprised of two energized rails - one above and one beneath, forming a complete electrical circuit

Transport vehicles will be deployed on these tracks. Each vehicle:

- ...Is less than 4 feet long

- ...Can hold approximately 20 pounds, the equivalent of an typical one bag average grocery shopping trip, or 2 gallon jugs of milk

- ...Is equipped with a computerized component, where the destination is entered

- ...When a vehicle and the enclosed package arrive at it's the correct destination, the payload container entire vehicle is ejected from the system

- ...The recipient removes the package and either:

- ...Saves the container for when he/she needs to send something through the system

- ...Returns the container to its original location by selecting return function, possibly returning any packaging to be recycled

- ...Puts a new payload into the container, entering new address, and placing container into system for immediate outbound shipment.

Unlike standard rail systems, there is no need to set switches or "traffic signals" in advance. How are collisions avoided, then?

It is a system known as an "onboard switch", with distributed and cooperative vehicle traffic algorithms. The system upper limit on vehicles travelled per lane mile will be about *10 times greater* than the limit on ordinary cars on highways, and more than 100 times that of any pneumatic system. In simple words - the system is built in such a way that it prevents a



In terms of energy, this system uses **less than 1% of the energy** used by any other system.
Why? Because the vehicles:

- ...Are self-propelled

- ...Weigh significantly less than its payload

- ...Do not suffer extra rolling resistance (unlike road vehicles, which need to support their weight on rubber tires.)

- ...Don't have to fly or navigate unknown environment

There are a few analogies to help explain how the CargoFish system works.

It will be like a capillary blood system for each neighborhood, with each vehicle and packet acting like a red blood cell.

The system resembles binary digital computer design, where any line can diverge from one into two, and any two lines may merge into one.

As the tagline says, it is a “physical internet”. Each “pilot” location is like one computer or internal network. The interconnection of several local systems will be like the Internet.

